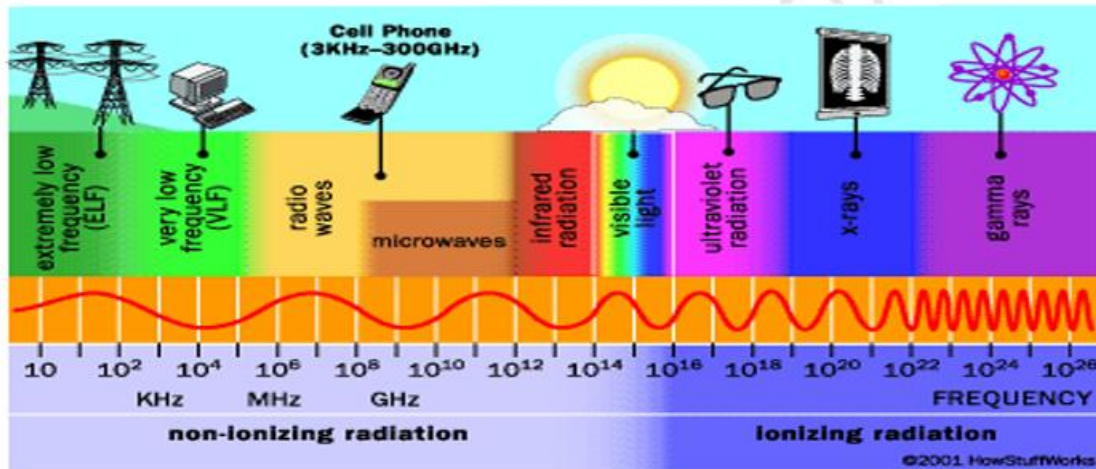


Radiology

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Hazard and Protection

Types of Radiation



Radiation is broadly classified into two main types:

Non-Ionizing Radiation: This is radiation that does not possess sufficient energy to dislodge orbital electrons. Examples include microwaves, ultraviolet (UV) light, lasers, radio waves, and infrared light.

Ionizing Radiation: This radiation has enough energy to dislodge orbital electrons from an atom. Examples include alpha particles, beta particles, neutrons, X-rays, and gamma rays.



Radiation biology is the specific study of how ionizing radiation affects living systems.

How Ionizing Radiation Affects Cells

The fundamental effect of ionizing radiation is **ionization**.

Mechanism: When an X-ray photon strikes an electron in an atom of living tissue, it displaces the electron. This leaves the atom electrically unbalanced, or "ionized". This ionized atom then seeks stability by reacting chemically, which forms new chemicals and alters the cell.

Basic Effects: The primary results of ionization are molecular alteration and the creation of new chemicals.

Timeline: This initial interaction at the electron level happens within 10-13 seconds of exposure. This leads to molecular changes within seconds to hours, which can persist for hours, decades, or even generations and may result in cell injury or death.

This cellular damage occurs through two main pathways:

Direct Effect: The energy from a photon or secondary electron directly ionizes biologic macromolecules (like DNA). In this case, all exposed cells in a specific area are altered directly by the ionization.

Indirect Effect: This occurs when new chemicals, resulting from ionization, are incompatible with body tissues. The most common example is the **radiolysis of water**. When a photon is absorbed by water in an organism, it ionizes the water molecules. This process forms free radicals, such as hydrogen peroxide (H_2O_2), which are toxins that cause cellular dysfunction. Radiation can also indirectly alter hormones, enzymes, and other body secretions, making them ineffective.

Stochastic vs. Deterministic Effects

The biological effects of radiation are categorized as either stochastic or deterministic.

Comparison of Stochastic and Deterministic Effects of Radiation

	Stochastic Effects	Deterministic Effects
Caused by	Sublethal DNA damage	Cell killing
Threshold dose	No There is no minimum threshold dose. Effect can be caused by any dose of radiation	Yes Effect occurs only when the threshold dose is exceeded
Severity of clinical effects and dose	Severity of clinical effects is independent of dose; all-or-none response—an individual either manifests effect or does not	Severity of clinical effects is proportional to dose; the higher the dose, the more severe the effect
Relationship between dose and effect	Frequency of effect proportional to dose; the higher the dose, the higher the risk of manifesting the effect	Probability of effect independent of dose; most individuals manifest effect when threshold dose is exceeded
Caused by doses used in diagnostic radiology	Yes	No
Examples	Radiation-induced cancer	Osteoradionecrosis
	Heritable effects	Radiation-induced cataract formation
	Radiation-induced skin cancer	Radiation-induced skin burns

Factors Affecting Biological Response

Radio-sensitivity of Tissues

A tissue's radiosensitivity is measured by its response to irradiation.

Cell Sensitivity: Cells are most sensitive to radiation when they are immature, undifferentiated, and rapidly dividing. They become less sensitive as they mature and specialize.

Oxygen Effect: Cells that are more oxygenated are more susceptible to radiation damage.

Tissue Ranking (High to Low Sensitivity):

- Highly sensitive tissues include lymphoid organs, blood-forming tissues (bone marrow), intestines, stem cells, lymphocytes, and reproductive cells.
- Tissues with intermediate sensitivity include young/growing bone and cartilage, glandular tissue, salivary glands, kidney, liver, and lungs.

- Tissues with low sensitivity include skin, muscle, and the optic lens.
- The least affected tissues are nerve tissue and adult bone.

Modifying Factors

The response of tissues to radiation depends on several factors:

Dose: The severity of deterministic damage is dependent on the amount of radiation received.

Dose Rate: This is the rate of exposure. A high dose rate (e.g., 5 Gy /min) causes more damage than the same total dose given at a low rate (e.g., 5 mGy /min).

Oxygen: Conducting irradiation with reduced oxygen (hypoxia) increases the radio-resistance of biological systems. The presence of oxygen allows for the formation of more toxic free radicals, like hydrogen peroxide.

Other Key Concepts

Short-term vs. Long-term Effects: Short-term effects are seen days or weeks after exposure, while long-term effects are seen months or years later.

Latent Period: This is the time between radiation exposure and the appearance of clinical symptoms. The more severe the dose, the shorter the latent period. This period can be as long as 25 years for some minimum doses.

Radiation Protection

The guiding law of radiation protection is the **ALARA principle**: "As Low As Reasonably Achievable". This means radiographs should only be taken at the minimum required dosage, and the benefit must be weighed against the radiation dose.

Protection of the Patient

The dentist is responsible for all aspects of safe radiation exposure. Methods include:

- 1) **Film Speed:** Using faster film requires less radiation and shorter exposure times.
- 2) **Digital Radiography:** Can provide a dose reduction of up to 90% compared to E-speed film.
- 3) **Collimation:** Using a collimating device, especially a rectangular one, prevents unnecessary beam divergence.
- 4) **Filtration:** This absorbs long-wavelength (soft) X-rays that have no diagnostic value.

- 5) **Distance (Cones):** Using a long cone (e.g., 16 inches) increases the tube-to-film distance, which, when paired with a higher kV, reduces the radiation dose absorbed by the skin.
- 6) **Technique:** Proper film placement and angulation are important to prevent retakes. Correct exposure times must be used (e.g., children and elderly patients may require a 50% reduction).
- 7) **Protective Gear:** A lead apron should be used to cover the patient's chest and reproductive areas. A thyroid collar is used to protect the thyroid gland. This is especially important for pregnant patients and young adults.
- 8) **Other Methods:** Using intensifying screens for extraoral films and lining the open cylinder with lead foil can eliminate scattered radiation.

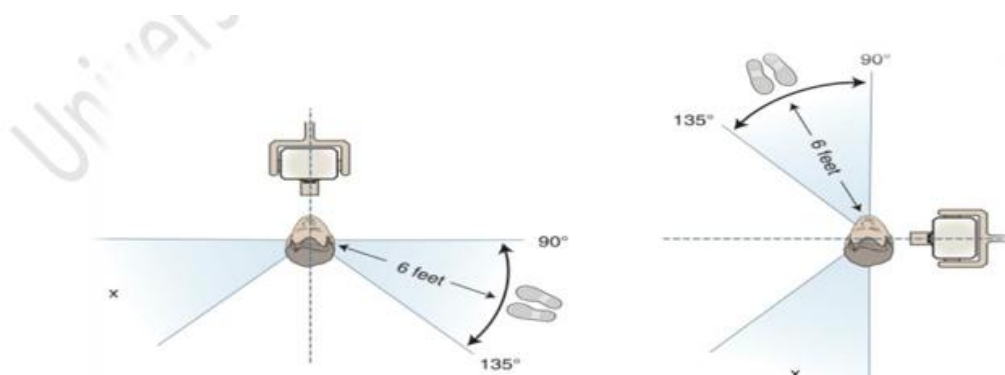
Protection of the Operator

The operator receives secondary radiation, which can be scattered from the patient, from the primary beam, or from tube head leakage. The whole-body radiation limit for workers should not exceed 5 rem per year.

To minimize exposure, the operator should use three methods:

Distance: The operator should stand at least **6 feet** away from the X-ray source. The intensity of radiation follows the inverse square law.

Position: The operator must stand at an angle of **90 to 135 degree** to the central ray of the X-ray beam. This is the area with the least scattered radiation. Standing behind the patient also helps, as the patient's head absorbs scatter.



Barrier: This is the most effective safety method. The operator should stand behind a barrier made of lead, steel, concrete, or barium plaster.

Dose Monitoring and Examples

Film Badges: Operators (radiographers and dentists) wear film badges to monitor their dose. It is a plastic frame with filters and radiographic film that provides a permanent record of the dose received. It is typically worn for 1-3 months before being processed.



Typical Effective Doses: The following table shows median effective doses for common radiographic exams:

Typical Effective Dose From Radiographic Examinations

Examination	Median Effective Dose	Equivalent Background Exposure ^a
Intraoral^b		
Rectangular collimation		
Posterior bite-wings: PSP or F-speed film	5 μ Sv	0.6 day
Full-mouth: PSP or F-speed film	40 μ Sv	5 days
Full-mouth: CCD sensor (estimated)	20 μ Sv	2.5 days
Round collimation		
Full-mouth: D-speed film	400 μ Sv	48 days
Full-mouth: PSP or F-speed film	200 μ Sv	24 days
Full-mouth: CCD sensor (estimated)	100 μ Sv	12 days
Extraoral		
Panoramic ^b	20 μ Sv	2.5 days
Cephalometric ^b	5 μ Sv	0.6 day
Chest ^c	100 μ Sv	12 days
Cone beam CT ^b		
Small field of view (<6 cm)	50 μ Sv	6 days
Medium field of view (dentoalveolar, full arch)	100 μ Sv	12 days
Large field of view (craniofacial)	120 μ Sv	15 days
Multidetector CT		
Maxillofacial ^b	650 μ Sv	2 months
Head ^c	2 mSv	8 months
Chest ^c	7 mSv	2 years
Abdomen and pelvis, with and without contrast ^c	20 mSv	7 years